

## Derivations of Rectifier

Electronics Engineering

Prepared by:- Mudit Saxena & Arpita Johri

# PARAMETERS

Average or DC values of Output Voltage & Load Current

## HALF WAVE RECTIFIER

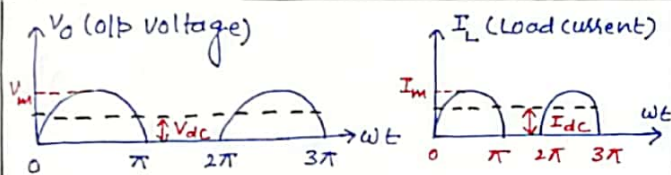


Fig (a) o/p voltage waveform For Half Wave Rectifier

Fig (b) Load current waveform For Half Wave Rectifier

⇒ Mathematically o/p voltage may be given as:-

$$V_o = V_m \sin \omega t, \text{ For } 0 \leq \omega t \leq \pi$$

$$= 0, \text{ For } \pi \leq \omega t \leq 2\pi \quad \text{--- (1)}$$

We know

$$V_{avg} \text{ or } V_{dc} = \frac{\text{Area under the Curve over Full cycle}}{\text{Base}} \quad \text{--- (A)}$$

but

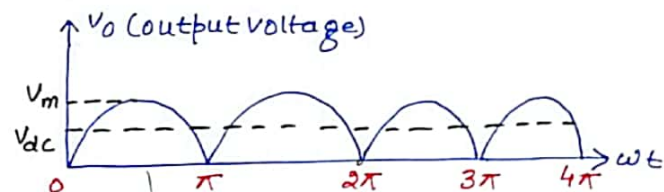
$$\text{Area} = \int_0^{2\pi} V_o d(\omega t) = \int_0^{\pi} V_o d(\omega t) + \int_{\pi}^{2\pi} V_o d(\omega t) \quad \text{--- (2)}$$

From eq<sup>n</sup> (1), put the value of  $V_o$  in eq<sup>n</sup> (2):-

$$\text{Area} = \int_0^{\pi} V_m \sin \omega t d(\omega t) + \int_{\pi}^{2\pi} 0 d(\omega t)$$

## FULL WAVE RECTIFIER

(1)



Average or DC value of o/p voltage

⇒ Mathematically o/p voltage may be given as:-

$$V_o = V_m \sin \omega t, \text{ For } 0 \leq \omega t \leq \pi$$

$$= -V_m \sin \omega t, \text{ For } \pi \leq \omega t \leq 2\pi$$

⇒ (-) sign indicates that during second half cycle, the sine wave is inverted

⇒ We know:-

$$V_{dc} = V_{avg} = \frac{\text{Area under the Curve over Full cycle}}{\text{Base}}$$

But

$$\text{Area} = \int_0^{2\pi} V_o d(\omega t) = \int_0^{\pi} V_o d(\omega t) + \int_{\pi}^{2\pi} V_o d(\omega t)$$

### HNR

$$\begin{aligned} A_{\text{rea}} &= V_m [-\cos \omega t]_0^\pi + 0 \\ &= V_m [-\cos \pi - (-\cos 0)] \\ &= V_m [1+1] \end{aligned}$$

$$A_{\text{rea}} = 2V_m$$

From eqn (A), we get :-

$$V_{\text{dc}} \text{ or } V_{\text{avg}} = \frac{2V_m}{2\pi} = \frac{V_m}{\pi} = 0.318V_m$$

⇒ Thus dc value of o/p voltage is 31.8% of the maximum A-C i/p voltage.

⇒ Similarly average or dc value of load current may be calculated as follows:-

$$\Rightarrow I_{\text{avg}} = I_{\text{dc}} = \frac{V_{\text{dc}}}{R_L}$$

$$\text{but } V_{\text{dc}} = \frac{V_m}{\pi} \text{ (above calculated)}$$

$$\Rightarrow \text{so } I_{\text{dc}} = \frac{V_m}{\pi R_L} \text{ but } \frac{V_m}{R_L} = I_m$$

### FNR

$$\begin{aligned} \Rightarrow A_{\text{rea}} &= \int_0^\pi V_m \sin \omega t d(\omega t) + \int_\pi^{2\pi} -V_m \sin \omega t d(\omega t) \\ &= [-V_m \cos \omega t]_0^\pi + [V_m \cos \omega t]_\pi^{2\pi} \\ &= V_m [-\cos \pi + \cos 0] + V_m [\cos 2\pi - \cos \pi] \\ &= V_m [1+1+1+1] \end{aligned}$$

$$A_{\text{rea}} = 4V_m$$

$$\text{so } V_{\text{dc}} = V_{\text{avg}} = \frac{A_{\text{rea}}}{\text{Base}} = \frac{4V_m}{2\pi} = \frac{2V_m}{\pi}$$

$$V_{\text{dc}} = V_{\text{avg}} = 0.636V_m$$

⇒ Thus dc value of o/p voltage is 63.6% of max. A-C input voltage.

⇒ Similarly avg or dc value of load current may be calculated as follows:-

$$I_{\text{avg}} = I_{\text{dc}} = \frac{V_{\text{dc}}}{R_L} \Rightarrow \text{but } V_{\text{dc}} = \frac{2V_m}{\pi} \text{ (above)}$$

### HWR

$$\Rightarrow I_{dc} = I_{avg} = \frac{I_m}{\pi}$$

$$I_{dc} = I_{avg} = 0.318 I_m$$

$\Rightarrow$  d.c value of Load current is 31.8% of max. value of load current

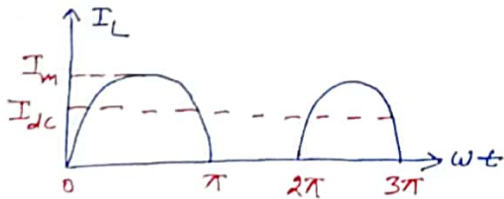
We know

Ripple factor  $k = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} \rightarrow \textcircled{1}$

$\Rightarrow$  For Half wave rectifier, we know

$$\Rightarrow I_{dc} = \frac{I_m}{\pi} \text{ (calculated above)}$$

$$\text{but } I_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} I_L^2 d(\omega t)}$$



### FWR

$$I_{dc} = \frac{2V_m}{\pi R_L} \text{ but } \frac{V_m}{R_L} = I_m$$

$$\Rightarrow I_{avg} = I_{dc} = \frac{2I_m}{\pi}$$

$$I_{avg} = I_{dc} = 0.636 I_m$$

$\Rightarrow$  DC value of Load current is 63.6% of max. value of load current

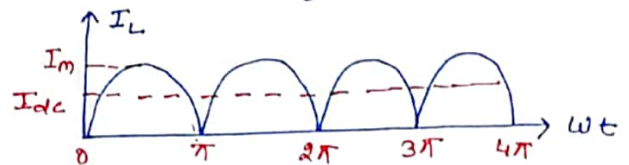
We know

$$k = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} \rightarrow \textcircled{1}$$

$\Rightarrow$  For Full wave rectifier, we know

$$I_{dc} = \frac{2I_m}{\pi} \text{ (above)}$$

$$\text{but } I_{rms} = \sqrt{\frac{1}{\pi} \int_0^{\pi} I_L^2 d(\omega t)}$$



### HWR

$$\text{but } I_L = I_m \sin \omega t \left\{ \begin{array}{l} \text{For } 0 \leq \omega t \leq \pi \\ = 0 \end{array} \right. \left\{ \begin{array}{l} \text{For } \pi \leq \omega t \leq 2\pi \end{array} \right.$$

$$I_{rms} = \sqrt{\frac{1}{2\pi} \left[ \int_0^\pi I_m^2 \sin^2 \omega t d(\omega t) + \int_\pi^{2\pi} 0 d(\omega t) \right]}$$

$$= \sqrt{\frac{1}{2\pi} \int_0^\pi I_m^2 \sin^2 \omega t d(\omega t)}$$

$$= \sqrt{\frac{I_m^2}{2\pi} \int_0^\pi \left( \frac{1 - \cos 2\omega t}{2} \right) d(\omega t)}$$

$$= \sqrt{\frac{I_m^2}{2\pi \times 2} \left[ \omega t - \frac{\sin 2\omega t}{2} \right]_0^\pi}$$

$$= \sqrt{\frac{I_m^2}{4\pi} \left[ (\pi - 0) - (0 - 0) \right]}$$

$$= \sqrt{\frac{I_m^2}{4}}$$

$$I_{rms} = \frac{I_m}{2}$$

### FWR

$$\text{but } I_L = I_m \sin \omega t, \text{ For } 0 \leq \omega t \leq \pi$$

Therefore

$$I_{rms} = \sqrt{\frac{1}{\pi} \int_0^\pi I_m^2 \sin^2 \omega t d(\omega t)}$$

$$= \sqrt{\frac{I_m^2}{\pi} \int_0^\pi \frac{(1 - \cos 2\omega t)}{2} d(\omega t)}$$

$$= \sqrt{\frac{I_m^2}{\pi} \left[ \frac{\omega t}{2} - \frac{\sin 2\omega t}{4} \right]_0^\pi}$$

$$= \sqrt{\frac{I_m^2}{\pi} \times \frac{\pi}{2}}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

but the value of  $I_{rms}$  &  $I_{dc}$  in eq<sup>n</sup> (1) :-

HWR

⇒ put the value of  $I_{dc}$  &  $I_{rms}$  in eqn ①

ripple factor  $r = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$

$$r = \sqrt{\left[\frac{(I_m/2)}{(I_m/\pi)}\right]^2 - 1}$$

$$r = 1.21$$

FWR

⑤

$$r = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$$

$$= \sqrt{\left(\frac{I_m/\sqrt{2}}{2I_m/\pi}\right)^2 - 1}$$

$$= \sqrt{\frac{\pi^2}{8} - 1}$$

$$r = 0.482$$
 which is much smaller than

Half wave rectifier, Due to this reason, Full wave rectifier is widely used.

③

Efficiency  
of Rectifier

We know

$$\text{Rectifier Efficiency } (\eta) = \frac{\text{dc Power Delivered}}{\text{A.C Input power}}$$

$$\eta = \frac{P_{dc}}{P_{ac}}$$

⇒ but

$$P_{dc} = I_{dc}^2 \times R_L$$

$$P_{ac} = I_{rms}^2 (R_L + R_F)$$

Where

$R_L \rightarrow$  Load Resistance

$R_F \rightarrow$  Forward Resistance of diode

Note:- 1) Ripple factor should be as low as possible

2) Ripple factor indicates that how close the rectified output is to the pure ideal dc voltage waveform.

HWR

⇒ put the value of  $I_{dc}$  &  $I_{rms}$  in eqn ①

ripple factor  $r = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$

$$r = \sqrt{\left[\frac{(I_m/2)}{(I_m/\pi)}\right]^2 - 1}$$

$$r = 1.21$$

FWR

⑤

$$r = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$$

$$= \sqrt{\left(\frac{I_m/\sqrt{2}}{2I_m/\pi}\right)^2 - 1}$$

$$= \sqrt{\frac{\pi^2}{8} - 1}$$

$$r = 0.482$$
 which is much smaller than

Half wave rectifier, Due to this reason, Full wave rectifier is widely used.

③

Efficiency  
of Rectifier

We know

$$\text{Rectifier Efficiency } (\eta) = \frac{\text{dc Power Delivered}}{\text{A.C Input power}}$$

$$\eta = \frac{P_{dc}}{P_{ac}}$$

⇒ but

$$\left. \begin{aligned} P_{dc} &= I_{dc}^2 \times R_L \\ P_{ac} &= I_{rms}^2 (R_L + R_F) \end{aligned} \right\}$$

Where

$R_L \rightarrow$  Load Resistance

$R_F \rightarrow$  Forward Resistance of diode

Note:- 1) Ripple factor should be as low as possible

2) Ripple factor indicates that how close the rectified output is to the pure ideal dc voltage waveform.

### HWR

$$\eta = \frac{I_{dc}^2 \times R_L}{I_{rms}^2 \times (R_L + R_F)} \rightarrow (A)$$

For Half Wave Rectifier

We know

$$\left. \begin{aligned} I_{dc} &= \frac{I_m}{\pi} \\ I_{rms} &= \frac{I_m}{2} \end{aligned} \right\}$$

$\Rightarrow$  put the values in eq<sup>n</sup> (A)

$$\eta = \frac{\left(\frac{I_m}{\pi}\right)^2 \times R_L}{\left(\frac{I_m}{2}\right)^2 \times (R_L + R_F)}$$

$$= \frac{4}{\pi^2} \times \frac{R_L}{(R_L + R_F)}$$

$$\eta = \frac{0.406}{1 + \frac{R_F}{R_L}}, \text{ Efficiency will be max, when } R_L \gg R_F$$

$$\Rightarrow \eta_{max} = 0.406 = 40.6\%$$

### FWR

(6)

For Full Wave Rectifier

We know

$$\left. \begin{aligned} I_{dc} &= \frac{2I_m}{\pi} \\ I_{rms} &= \frac{I_m}{\sqrt{2}} \end{aligned} \right\}$$

$\Rightarrow$  put the values in eq<sup>n</sup> (A):

$$\eta = \frac{\left(\frac{2I_m}{\pi}\right)^2 \times R_L}{\left(\frac{I_m}{\sqrt{2}}\right)^2 \times (R_L + R_F)} = \frac{8}{\pi^2} \times \frac{R_L}{R_L + R_F}$$

$$\eta = \frac{0.812}{1 + \frac{R_F}{R_L}}$$

$\Rightarrow$  Efficiency will be max, when  $R_L \gg R_F$

$$\eta_{max} = 0.812 = 81.2\% \text{ which is twice as}$$

that of HWR. Hence a FWR is twice as effective as a HWR.

④

Peak Inverse Voltage (PIV)

HWR

PIV is the maximum amount of voltage that a diode has to withstand when it is operated in reverse bias.

PIV = Vm -> For Half wave

⑤

Transformer Utilization Factor (TUF)

TUF = DC Power Delivered to the Load / AC power rating of transformer secondary

TUF = (Vdc Idc) / (Vrms Irms)

For Half wave Rectifier :-

TUF = (Vdc Idc) / (Vrms Irms)

FWR

⑦

For Full wave < Centre tapped Bridge Rectifier

=> For Centre tapped :-

PIV = 2Vm

=> For Bridge :-

PIV = Vm

=> For Full wave Rectifier :-

TUF = (TUF(Primary) + TUF(secondary)) / 2

TUF = (0.574 + 0.812) / 2

TUF = 0.693 = 69.3%

### Half-wave Rectifier

### Full wave Rectifier

(8)

$$\Rightarrow \text{but } \left. \begin{array}{l} V_{dc} = V_m / \pi \\ I_{dc} = I_m / \pi \\ V_{rms} = V_m / \sqrt{2} \\ I_{rms} = I_m / 2 \end{array} \right\} \text{ For Half Wave Rectifier}$$

$\Rightarrow$  put these values, we get:-

$$TUF = \frac{[V_m / \pi]}{[V_m / \sqrt{2}]} \cdot \frac{[I_m / \pi]}{[I_m / 2]} = \frac{2\sqrt{2}}{\pi^2}$$

$$TUF = 0.287 = 28.7\%$$

$\Rightarrow$  TUF should be as high as possible.

$\Rightarrow$  TUF For Half wave Rectifier = 28.7%

indicates that the transformer is being utilized to only 28.7% of its full capacity